Last but not least, let's see how the logs' backend OpenSearch is configured (in the Docker Compose file in the following listing) in a single-node setup along with the frontend/UI, OpenSearch Dashboards.

Listing 4.3 Docker Compose configuration snippet for OpenSearch (and Dashboards)

```
opensearch-node1:
    image: opensearchproject/opensearch:2.1.0
    container_name: opensearch-node1
    environment:
      - cluster.name=opensearch-cluster
      - node.name=opensearch-node1
      - bootstrap.memory_lock=true
      - "OPENSEARCH_JAVA_OPTS=-Xms512m -Xmx512m"
      - "DISABLE_INSTALL_DEMO_CONFIG=true"
      - "DISABLE_SECURITY_PLUGIN=true"
      - "discovery.type=single-node"
    volumes:
      - opensearch-data1:/usr/share/opensearch/data
    ports:
      - 9200:9200

  opensearch-dashboards:
    image: opensearchproject/opensearch-dashboards:2.1.0
    container_name: opensearch-dashboards
    ports:
      - 5601:5601
    environment:
      - 'OPENSEARCH_HOSTS=["http://opensearch-node1:9200"]'
      - "DISABLE_SECURITY_DASHBOARDS_PLUGIN=true"
```

Exposes port 9200 so that we can ingest log lines (from Fluent Bit)

Exposes port 5601 to make OpenSearch Dashboards (frontend/UI) accessible from the host machine

Configures OpenSearch Dashboards (frontend) to use OpenSearch (backend for logs)

At this point, we've effectively implemented the logging pipeline shown in figure 4.1. Now, let's have a closer look at the Fluent Bit configuration itself. As depicted in listing 4.4, I'm using the Forward (https://docs.fluentbit.io/manual/pipeline/inputs/forward) input plug-in in Fluent Bit, taking in the log stream from Fluentd (from the Docker runtime) and, via the OpenSearch output section, configuring Fluent Bit to route the log lines to OpenSearch.

Listing 4.4 Fluent Bit configuration

Here, we're configuring the Forward plug-in capable of receiving log items from Fluentd (from the Docker log driver).

The Forward plug-in will listen on all network interfaces (that is, 0.0.0.0).

The default port for the Forward plug-in (that Fluentd connects to)

```
[INPUT]
    Name forward
    Listen 0.0.0.0
    Port 24224
```